

Topics on the Edge

Pushing the fundamental limits of federated learning on the mobile edge

- Federated Learning and Mobile Edge Computing
- Hierarchical FL
- Compute-aware Client Selection

Federated Learning and the Mobile Edge

The Mobile Edge

“Enduring Challenges” of Pervasive Computing

- Resource poverty
- Communication Uncertainty
- Finite Energy
- *Multi-modal interaction*
- *Scarce user attention*
- Lower privacy, security, robustness

Federated Learning: Natural Advantages

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- Client Selection
- Federated Training

Federated Learning: Natural Disadvantages

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There will always be a strong case for centralization!

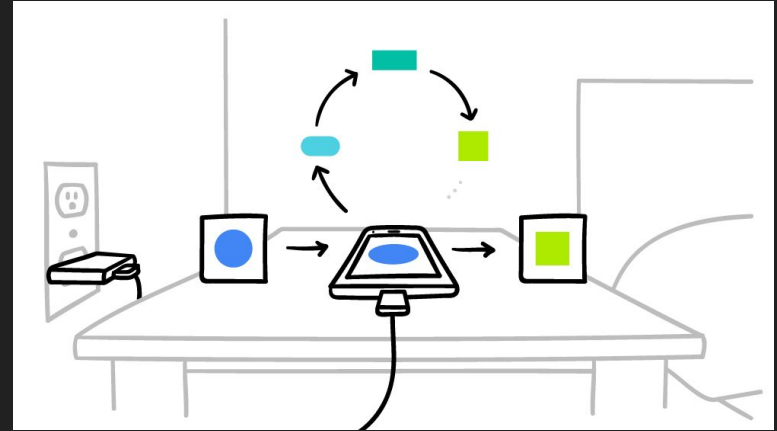
Current Federated Learning

Current Applications:

- Next word prediction
- Recommender systems

Not very performance sensitive

- Recruit only plugged-in, wifi-connected clients
- Research focus is mostly on accuracy, data heterogeneity, etc
- Does not hit the fundamental limits of mobile computing

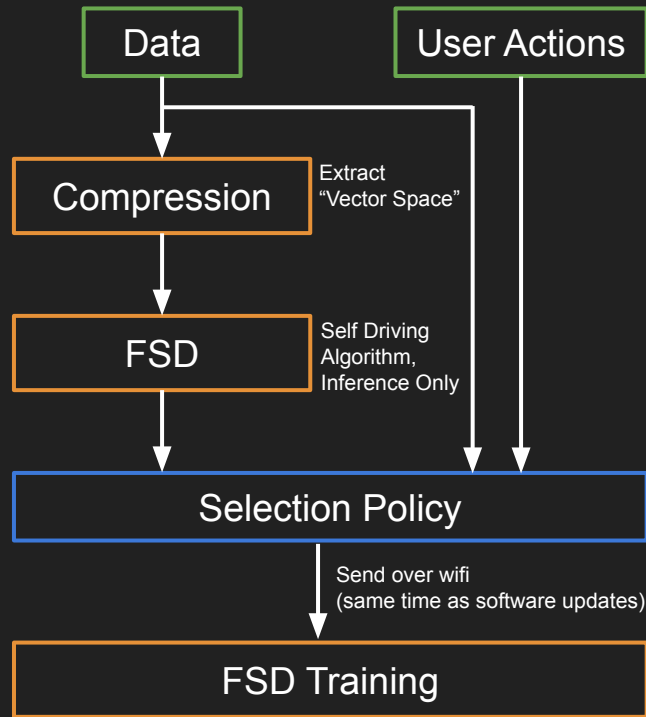


[Google - Federated Learning for Google Keyboard](#)

Why Mobile Edge?

Why (Not) Mobile Edge?

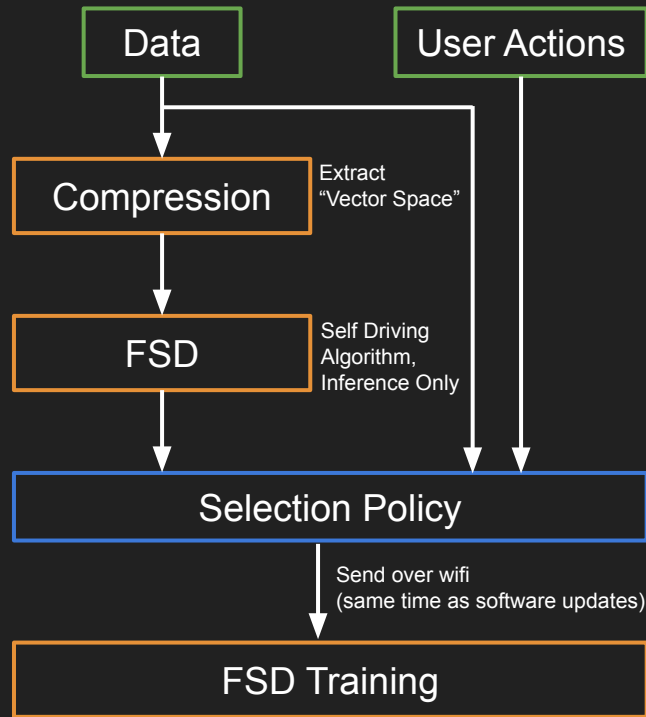
Tesla Full Self Driving Training



Why (Not) Mobile Edge?

Tesla Full Self Driving Training

- Edge training is hard/expensive
- Users don't know about privacy
- Users don't care about privacy



No articles about privacy on the first page!

People are only mildly concerned: [Self-Driving Cars and Data Collection: Privacy Perceptions of Networked Autonomous Vehicles](#) (2017, Lujó Bauer's group)

Google tesla autopilot training data

Q All News Videos Images Shopping More Tools

About 12,800 results (0.27 seconds)

FutureCar
Toyota's Woven Planet is Now 'Training' its Self-Driving ...
Toyota's Woven Planet is Now 'Training' its Self-Driving Development Vehicles Using Data From Low-Cost Cameras, an Approach Used by Tesla.
2 weeks ago

Automotive IQ
IQ News: Toyota and Tesla's Self-Driving Approach Leads to more Affordable AVs | Automotive IQ
Tesla has proven that collecting data with self-driving cars through ... of training data to support fully self-driving functionalities...
2 weeks ago

carandbike
Elon Musk Says In TED Video FSD Beta Has 100,000 Users
... to the amount of training data it was gathering via real world usage. ... FSD is Tesla's successor to AutoPilot which has been largely...
12 hours ago

Not a Tesla App
2022.4.5.20 Official Tesla Release Notes - Software Updates
For maximum safety and accountability, use of Full Self-Driving (Beta) will ... the data size of the next-gen autolabeler, training network...
3 weeks ago

Not a Tesla App
2022.4.5.21 Official Tesla Release Notes - Software Updates
A disengagement is when the Autopilot system disengages for the remainder of ... the data size of the next-gen autolabeler, training network...
2 weeks ago

Analytics Insight
Want to Land Your Dream ML Job at Tesla? Here's How?
The Simulation team realizes these goals through generating synthetic datasets for neural network training, building tools that enable Autopilot...
1 week ago

VentureBeat
What is autonomous AI? A guide for enterprises
Join AI and data leaders for insightful talks and exciting networking ... One of the developers of Tesla's autopilot software, for instance...
3 weeks ago

Google tesla autopilot training data

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About 404,000 results (0.44 seconds)

<https://towardsdatascience.com/tesla-ai-day-2021-revi...>
Tesla AI Day 2021 Review — Part 2: Training Data. How Does ...
Sep 23, 2021 — Tesla is combining manual labeling, auto labeling, and simulation to create real-world datasets for fully self-driving cars.

<https://towardsdatascience.com/teslas-deep-learning-at-...>
Tesla's Deep Learning at Scale: Using Billions of Miles to ...
May 7, 2019 — Tesla's advantage in training data implies an advantage in object detection, prediction, and path planning/driving policy.

People also ask

- Does Tesla Autopilot use deep learning?
- Is Tesla self-driving machine learning?
- Does Tesla Autopilot collect data?
- What programming language does Tesla Autopilot use?

<https://www.tesla.com> > ...
Artificial Intelligence & Autopilot | Tesla
Build AI training chips to power our Dojo system. Implement bleeding-edge technology from the smallest training nodes to the multi-die training tiles.
[Autopilot](#) [FSD Chip](#) [Tesla Australia](#)

Videos

Tesla FULL self driving explained by an engineer (with Elon ...
YouTube · CNET Highlights
Aug 19, 2021

10 key moments in this video

- From 02:59 Biological Visual Cortex Wiring
- From 04:57 Neural Network Backbone
- From 06:12 Detection Head
- From 09:48 Problem: Per-Camera Detection...
- From 11:1 Multi-Cam Vector Spac Predictions

Tesla AI Day
YouTube · Tesla
Aug 19, 2021

10 key moments in this video

Andrei Karpathy - AI for Full-Self Driving at Tesla
YouTube · Matroid

Why Mobile Edge?

- Privacy and Liability
 - Increased awareness of data privacy
 - GDPR and “[Personal Data Sovereignty](#)”
 - Liability to data breaches and the difficulty in obtaining [cyber insurance](#)

Why Mobile Edge?

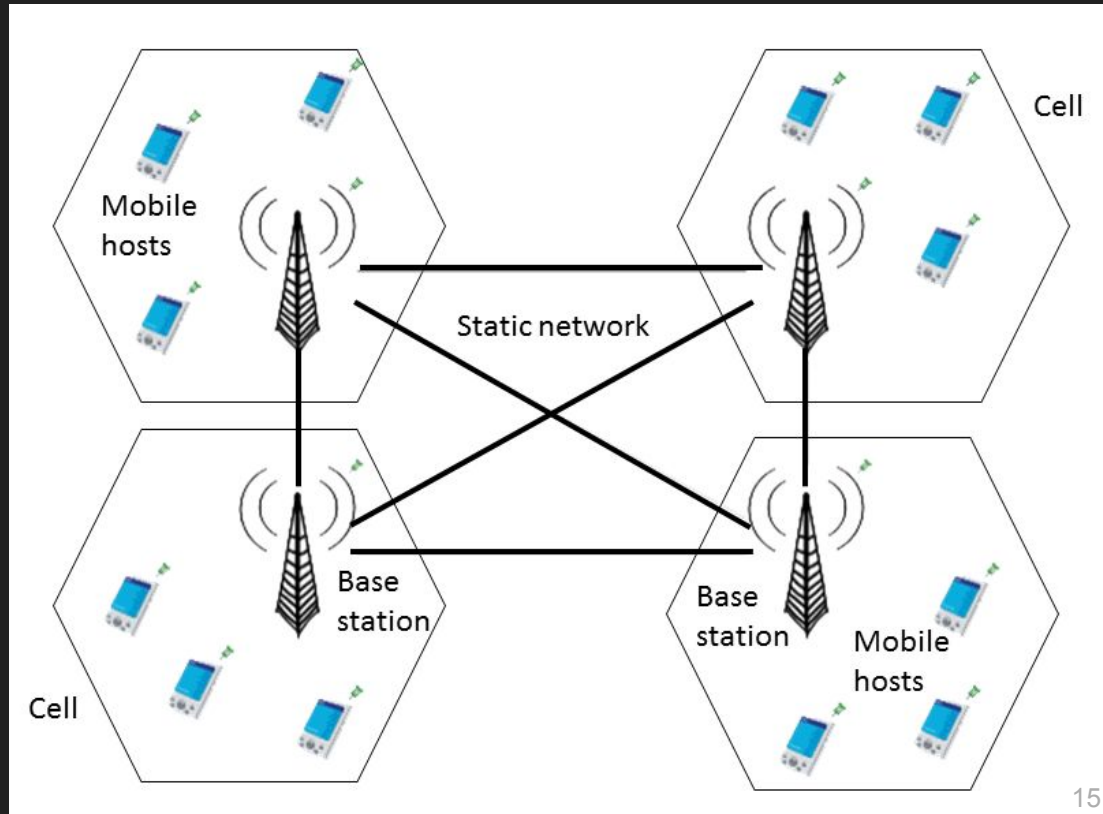
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- Communication Constraints
 - High data rate, i.e. video, LIDAR (easily 100s of gbps)
 - More domain-specific tasks where you need more data from each client

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- **Rapid Iteration**
 - Relative to training time
 - Without rapid iteration, none of these constraints matter!

Hierarchical FL

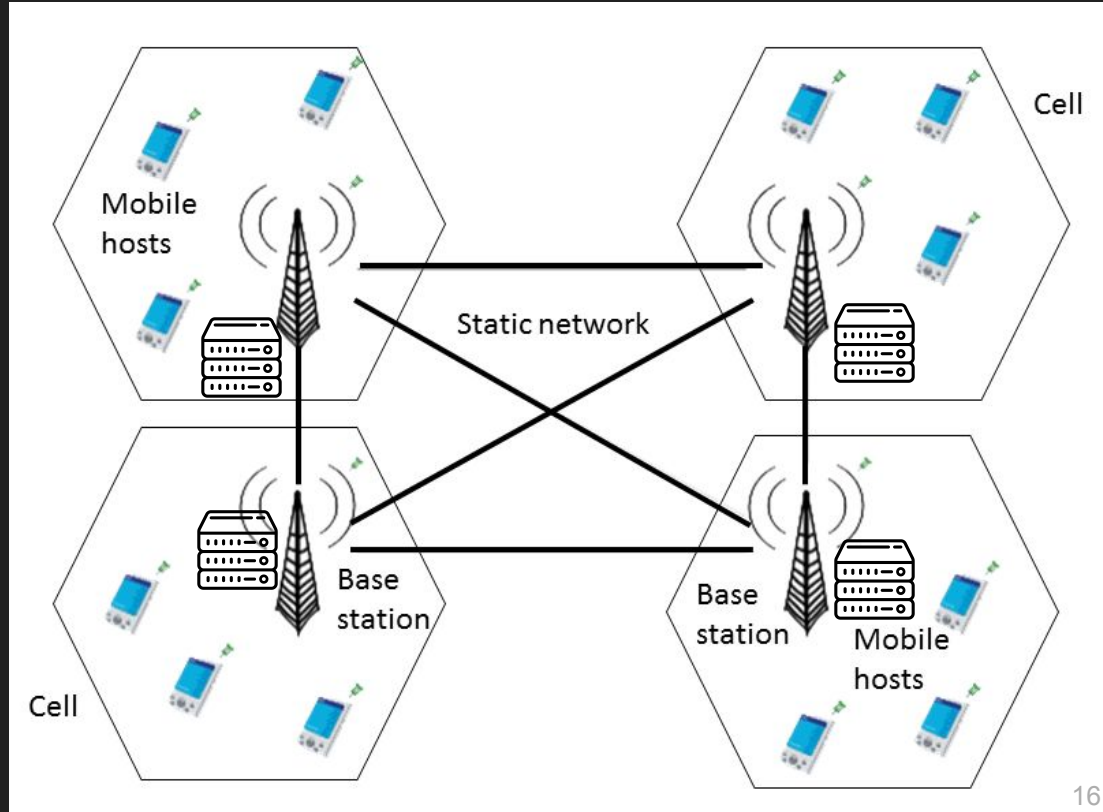
Physical Architecture



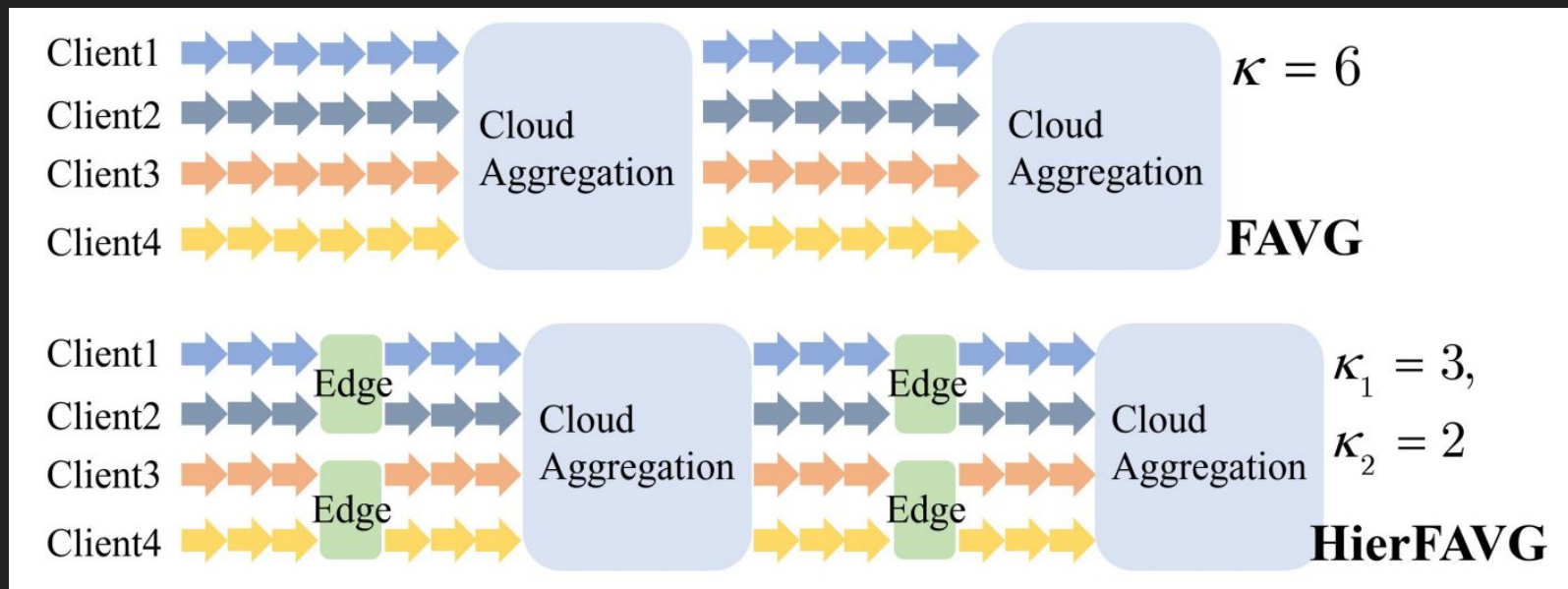
Just add Edge Servers!

Verizon 5G Edge

(It's an AWS virtual machine connected to the cellular network somewhere)



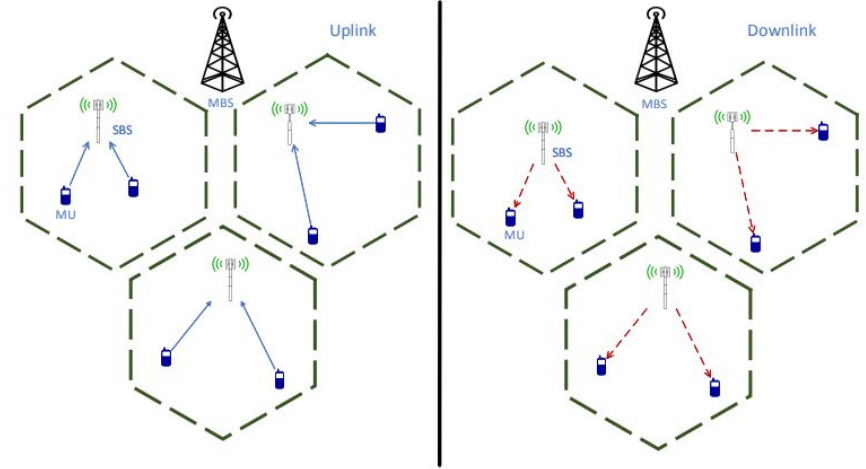
Hierarchical Federated Learning



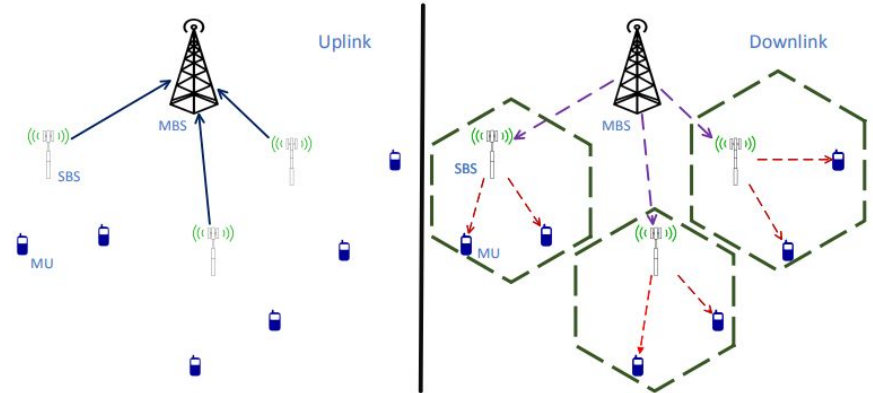
[Client-Edge-Cloud Hierarchical Federated Learning](#) (2019)

Hierarchical Federated Learning

Hierarchical Federated Learning
Across Heterogeneous Cellular
Networks (2019)

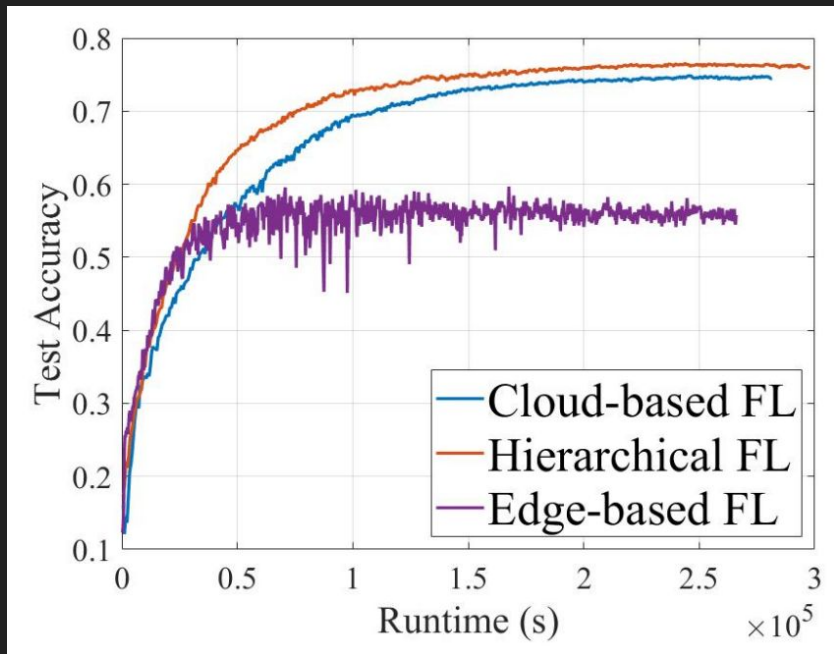


(a) Local gradient update

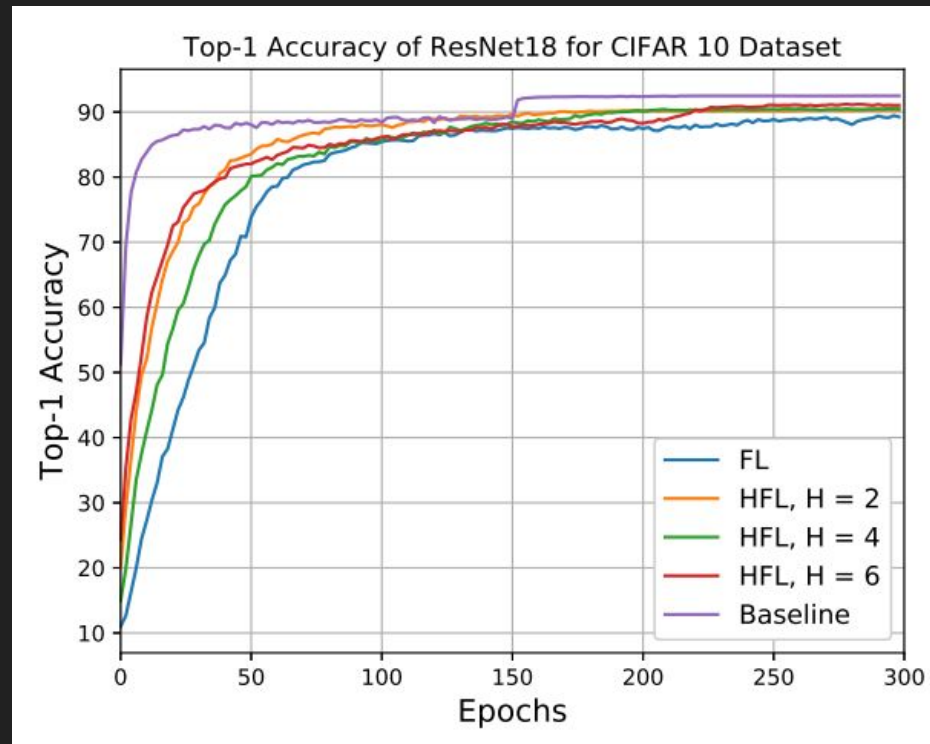


(b) Global model averaging

More Synchronization = Faster Convergence



*experiments use 50 and 28 clients, respectively



Hierarchical Federated Learning

Extensions:

- [Arbitrarily many levels](#) (2022)
- [Edge aggregation server selection and scheduling](#) (2020)

Possible Ideas:

- Per-cluster adaptation (number of steps, gradient compression, etc.)
- Cluster selection and client selection
- Mixed hierarchical, non-hierarchical FL

Why Not Now?

Hierarchical FL is “canonical,” but...

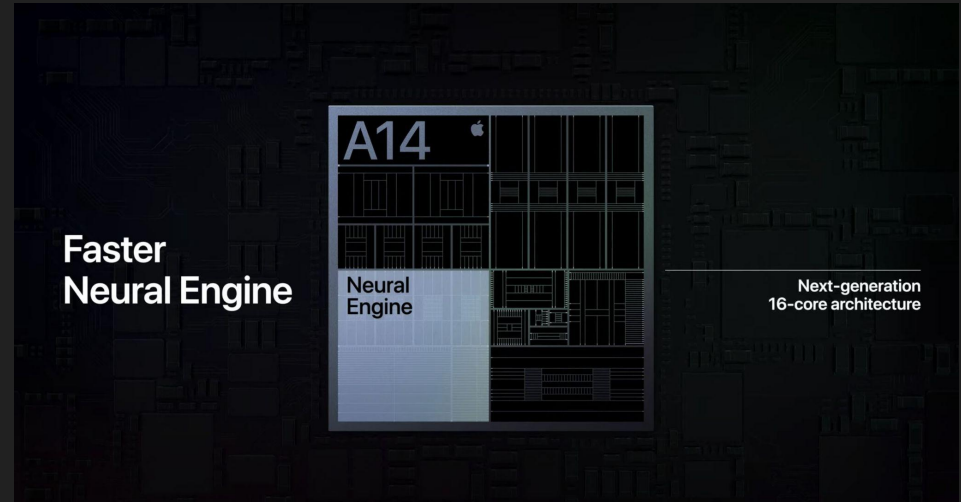
Why Not Now?

Hierarchical FL is “canonical,” but...

- Infrastructure is not fully there yet
- Edge server deployment is difficult
- Data is billed to the mobile user, not the developer. No discount for edge communication.

Compute-Aware Client Selection

Today's Server, Tomorrow's Edge



Compute-Aware Client Selection

Most obvious approach: pick all clients that will complete in time.

Client Selection for Federated Learning with Heterogeneous Resources in Mobile Edge, 2019

Algorithm 3 Client Selection in Protocol 2

Require: Index set of randomly selected clients $\mathbb{K}'$

```
1: Initialization  $\mathbb{S} \leftarrow \{\}$ ,  $T_{\mathbb{S}=\emptyset}^d \leftarrow 0$ ,  $\Theta \leftarrow 0$ 
2: while  $|\mathbb{K}'| > 0$  do
3:    $x \leftarrow \arg \max_{k \in \mathbb{K}'} \frac{1}{T_{\mathbb{S} \cup k}^d - T_{\mathbb{S}}^d + t_k^{\text{UL}} + \max\{0, t_k^{\text{UD}} - \Theta\}}$ 
4:   remove  $x$  from  $\mathbb{K}'$ 
5:    $\Theta' \leftarrow \Theta + t_x^{\text{UL}} + \max\{0, t_x^{\text{UD}} - \Theta\}$ 
6:    $t \leftarrow T_{\text{cs}} + T_{\mathbb{S} \cup x}^d + \Theta' + T_{\text{agg}}$ 
7:   if  $t < T_{\text{round}}$  then
8:      $\Theta \leftarrow \Theta'$ 
9:     add  $x$  to  $\mathbb{S}$ 
10:  end if
11: end while
12: return  $\mathbb{S}$ 
```

In summary, Client Selection is formulated by the following maximization problem with respect to $\mathbb{S}$:

$$\begin{aligned} \max_{\mathbb{S}} \quad & |\mathbb{S}| \\ \text{s.t.} \quad & T_{\text{round}} \geq T_{\text{cs}} + T_{\mathbb{S}}^d + \Theta_{|\mathbb{S}|} + T_{\text{agg}}. \end{aligned} \tag{4}$$

Compute-Aware Client Selection

[FedMCCS: Multicriteria Client Selection Model for Optimal IoT Federated Learning](#), 2021

B. Problem Formulation

We formulate our problem as a bilevel maximization with knapsack and other constraints as follows:

$$\max_{X_S} |X_S|$$

subject to

$$\begin{cases} \forall X_{f_z^i} \sum \text{Util}_{r \in \{\text{CPU}, \text{Memory}, \text{Energy}\}}^{X_{f_z}} < \text{Budget}_r^{X_{f_z}} [\text{co}_1] \\ \forall X_{f_z^i} \sum \left(T_d^{X_{f_z}} + \text{Util}_{r=T_{ud}}^{X_{f_z}} + T_{ul}^{X_{f_z}} \right) < T[\text{co}_2] \end{cases}$$

subject to

$$\max ER_{X_{f_z^i}} = \left[\frac{|X_{f_z} \cdot l_A|}{|X_{f_z} \cdot l_A| + |X_{f_z} \cdot l_N|} \times 100 \right] [\text{co}_3]. \quad (1)$$

Compute-Aware Client Selection

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Select as many clients as possible, such that:

(1) we do not exceed the resource budget

(2) we do not exceed the round time

(3) selection also maximizes clients with minority classes

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subject to

Percent of “abnormal” samples

$$\max ER_{X_{f_z}^i} = \left[\frac{|X_{f_z} \cdot l_A|}{|X_{f_z} \cdot l_A| + |X_{f_z} \cdot l_N|} \times 100 \right] [co_3]. \quad (1)$$

Compute-Aware Client Selection

Open questions:

- How do you select the budget and round time?
- How do you reconcile compute-aware selection with fairness if data is correlated to compute in hard-to-quantify ways?
- How do you know (i.e. predict) the resource usage ahead of time, especially if a device hasn't participated recently or has never participated?

Related work: [Runtime Performance Prediction for DL](#), 2021

Why Not Now?

- Under-studied (probably because it's really hard to study!)
- Performance-sensitive Federated Learning on the mobile edge not really used or needed yet
- Google: devices are limited in diversity, well-profiled in advance, and developers have root access

Conclusion

Hierarchical FL

- Obvious, simple, useful
- Very easy to implement in the lab
- Very hard to implement in the wild

Compute-aware Client Selection

- Logical, but not so simple
- Not needed until FL becomes more popular (especially by non-privileged parties)